EXPERIMENT 7

DRY RUN

Sample Input:

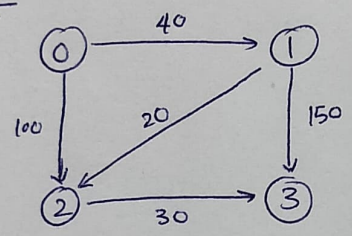
n=4 flights = [[0,1,40], [0,2,100], [1,2,20], [1,3,150], [2,3,30]]

src = 0

dst = 3

k=1

Graph:



cost = [∞, ∞, ∞, ∞]

cost [serc] = cost [0] = 0

cost = [0, ∞, ∞, ∞]

tmp = [0, ∞, ∞, ∞]

Iteration 0 (i=0 and i<=k)

Step	edge u→v	weight (w)	check if cost[u] != ∞ and cost[u] + w < tmp[v]	Update tmp[v] = cost[u] + w	tmp
0	0→1	40	cost[0] = 0 0+40 < ∞ True	tmp[1] = 0+40 =40	[0, 40, ∞, ∞]
1	0→2	100	cost[0] = 0 0+100 < ∞ True	tmp[2] = 0+100 =100	[0, 40, 100, ∞]
2	1→2	20	cost[1] = ∞ False	No update	[0, 40, 100, ∞]
3	1→3	150	cost[1] = ∞ False	No update	[0, 40, 100, ∞]
4	2→3	30	cost[2] = ∞ False	No update	[0, 40, 100, ∞]

Update cost = tmp

cost = [0, 40, 100, ∞]

<u>Iteration 1</u> (i=0 and i<=k)

step	edge $u \to v$	weight (w)	check if cost[u] != ∞ and cost[u] + w < tmp[v]	Update tmp[v] = cost[u] + w	tmp
0	0→1	40	cost[0] = 0 0+40 < 40 False	No update	[0, 40, 100, ∞]
1	0→2	100	cost[0] = 0 0+100 < 100 False	No update	[0, 40, 100, ∞]
2	1→2	20	cost[1] = 40 40+20 < 100 True	tmp[2] = 60	[0, 40, 60, ∞]
3	1→3	150	cost[1] = 40 40+150 < ∞ True	tmp[3] = 190	[0, 40, 60, 190]
4	2→3	30	cost[2] = 100 100+30 < 190 True	tmp[3] = 130	[0, 40, 60, 130]

Update cost = tmp

cost = [0, 40, 60, 130]

Answer: cost[dst] = cost[3] = 130